INVITED PAPER

Future E-Enabled Aircraft Communications and Security: The Next 20 Years and Beyond

This paper shows how the envisioned e-enabled aircraft play a central role in streamlining system modernization efforts; challenges, emerging solutions, and open problems are highlighted.

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ABSTRACT | Aircraft data communications and networking are key enablers for civilian air transportation systems to meet projected aviation demands of the next 20 years and beyond. In this paper, we show how the envisioned e-enabled aircraft plays a central role in streamlining system modernization efforts. We show why performance targets such as safety, security, capacity, efficiency, environmental benefit, travel comfort, and convenience will heavily depend on communications, networking and cyber-physical security capabilities of the e-enabled aircraft. The paper provides a comprehensive overview of the state-of-the-art research and standardization efforts. We highlight unique challenges, recent advances, and open problems in enhancing operations as well as certification of the future e-enabled aircraft.

KEYWORDS | Aviation cyber security; aviation security; cyber-physical systems; e-enabled aircraft

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I. INTRODUCTION

A. Modernization of Air Transportation

After thriving for a century, many civilian air transportation systems are aging today without the growth capacity to outpace future aviation demands [1]. For the next two decades and beyond, predictions show a multi-fold air traffic increase. Commercial and noncommercial aviation users demand higher efficiency and predictability in operations. Society mandates "green" savings in fossil fuel consumption, noise, and noxious gas emissions. Moreover, globalization and increasingly aging population indicate air travelers desire to connect personal devices for entertainment and healthcare in flight. Furthermore, risks from physical threats (e.g., bad weather, solar flares, human) are expected to be persistent, requiring protection of aviation information and physical assets at all times.

A prominent example of civil aviation's dire state is the world's biggest national airspace system in the United States, where one of the youngest air traffic infrastructure is 43 years old [1]. With an average of 7000 aircraft in the air at any given time, the system incurs annual delays of over a million hours that cost billions of dollars and burns 38 million gallons per day that emits over 12% of transportation CO_2.

Air transport system modernization efforts around the world, such as NextGen in the United States[1] and Single European Sky Air Traffic Management Research in Europe,[2] are renewing infrastructure and technology in

[1]http://www.jpdo.gov/
[2]http://www.eurocontrol.int/sesar

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Fig. 1. *E-enabled aircraft and civil aviation communities.*

space, air, and ground to meet future aviation demands. An overarching goal of these efforts is to enhance airspace system performance in terms of safety, security, capacity, and efficiency.

B. The E-Enabled Aircraft

A key vision in civil aviation modernization is the "e-enabled aircraft." Boeing 787 is a seminal example [2]. The e-enabled aircraft possesses advanced sensing, computing, communications, networking capabilities as well as onboard systems integration and software modules. It is envisioned to robustly participate as an intelligent node in a global network of air, satellite, ground systems, ensuring quality information correctly, and reliably reaches the right place at the right time for processing and decision making. Aviation stakeholders and society worldwide receive valuable services (see Fig. 1).

Future e-enabled aircraft must support more seamless, efficient, secure operations globally to help improve performance of future airspace systems that operate in an unforgiving operational environment. Towards this goal of enhancing aircraft operations, this paper addresses using data communications for building tight integration between onboard systems as well as between aircraft and off-board systems. Due to direct implications to airspace system performance, the paper focuses on the latter integration and how it enables close coordination of air traffic and timely refresh of aircraft digital content by net-enabled stakeholders.

C. Enhancing Aircraft Mobility in Air Traffic Systems

Future air traffic management (ATM)[3] systems benefit from aircraft capabilities that help with air traffic control goals such as interaircraft spacing reduction, automation and decentralization, and safe and secure flights that have optimal routes in terms of delay and costs. As seen later in Sections IV and V, compared to aging radar and legacy ATM infrastructures, global positioning system (GPS) [3], automatic dependent surveillance broadcast (ADS-B) [4] and Internet-protocol-based aeronautical telecommunication network (IP ATN) [5] can tightly integrate an aircraft with ground, air, and satellite systems in ATM. An onboard GPS sensor makes an aircraft interact with the global navigation satellite system for computing precise 4-D positions during flight. ADS-B broadcasts this position and the aircraft's identity allowing neighbor off-board systems to automatically, independently, safely, securely monitor the aircraft's flight route. IP ATN communicates voice and text between pilots, air traffic controllers, and airlines for flight-critical status updates and flight optimizing services.

D. Enhancing Aircraft Digital Content Delivery

Future aircraft will be increasingly rich in digital content due to the need for enhanced operations. For instance, as the role of aircraft software evolves to manage every aspect of flight, aircraft software criticality, size, and complexity grow exponentially [6]. Further, increasing sensor modalities for monitoring status of onboard physical assets—hardware (e.g., engine, structure, cable, computers), energy (e.g., fuel, power), cabin supply (e.g., life vests, food), passengers and crew (e.g., audio/video)—produce large volumes of data. Furthermore, passenger applications are increasingly multimedia intensive (e.g., massive entertainment databases) and critical (e.g., telemedicine, live events). Hence, timely refresh and retrieval of aircraft digital content by off-board systems of stakeholders, during flight as well as throughout aircraft lifecycle, is a vital need for future aircraft.

Legacy distribution of digital content via physical distribution of storage media, e.g., floppy and compact discs, and signed documents over bonded carriers is not sufficient. Tight integration between aircraft and ground systems via broadband IP networking offer digital content delivery to/from airlines and other stakeholders (see Section VI).

E. Security Considerations for Future Aviation

Tight integration within aircraft and between aircraft and off-board systems, however, raises new "cyber–physical" security considerations [8], mainly due to the risk to onboard integrated systems from remote and passenger "brought-in" devices and the higher dependence of flight on data communications. Cyber elements, i.e., computers and networks, have vulnerabilities to malicious disruption and misuse. Cyber threats must be assessed for their potential impact on expected physical behavior and performance gains of future aircraft and airspace systems. Furthermore, physical threats due to bad weather, icing, winds, solar flares, can be detrimental to cyber performance (e.g., GPS outage). Another challenge besides vulnerability assessment,

[3]Table 1 in Appendix has the abbreviations commonly used in this paper.

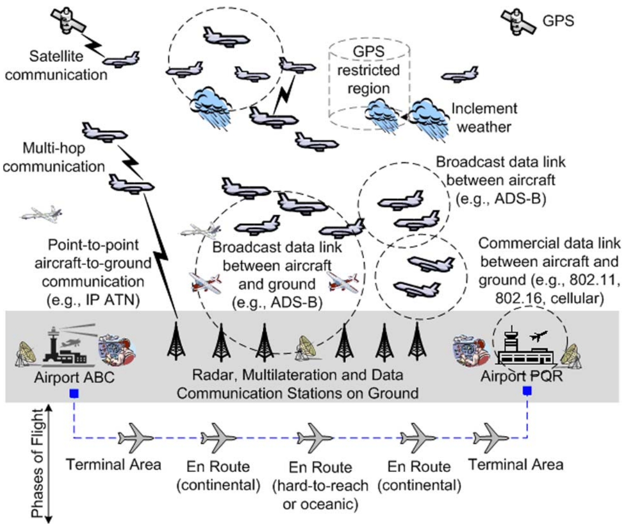

Fig. 2. ***Envisioned airspace system with future e-enabled aircraft.***

mitigation, exploit detection, and response is that aircraft must be certified to have minimal risk from cyber–physical threats. Most aviation standards do not yet cover cyber–physical threats. Community efforts have begun to bridge some of these gaps [6], [9], [10].

With a focus on security, this paper contributes research challenges, advances, and open problems in major aircraft communication capabilities, i.e., ADS-B, IP ATN, and digital content delivery, envisioned to be key players in the transformation of civil aviation in the next several decades.

F. Paper Outline

Section II describes the envisioned net-centric airspace system in which future e-enabled aircraft will need to enhance operations. Section III presents the envisioned e-enabled aircraft network employing tight integration. Section IV presents ADS-B and its security in greater detail due to its high criticality to future airspace systems performance [30]. Section V discusses IP ATN networking and security challenges. Section VI presents advances in secure aircraft digital content delivery to/from different stakeholders. Section VII discusses future aircraft certification process and emerging issues. Section VIII concludes the paper.

II. FUTURE AIRSPACE SYSTEM

Fig. 2 is a high-level illustration of a civilian airspace system with aircraft communications, envisioned for coming decades. It is extremely complex with heterogeneous types of future e-enabled aircraft, distributed infrastructures and systems-of-systems on the ground—all working together to safely, securely accommodate aviation efficiency demands, and grow airspace capacity in an unforgiving, unpredictable operational environment. Airspace users include general aviation, airlines, freight, and unmanned aircraft operators.

Aircraft-to-ground and aircraft-to-aircraft communications are used to monitor and control movement of traffic in the airspace system. As illustrated, communications over ADS-B data link and IP ATN (terrestrial or satellite radio link) help aircraft navigate in 4-D routes. Each route typically begins and ends at an airport and has terminal area and en-route flight phases. Air traffic controllers fuse surveillance data from multiple sensors on the ground, including primary and secondary surveillance radar, multilateration stations, and ADS-B data, to monitor position, movement, and conformance of aircraft on these routes. Any tactical or strategic change in flight route and schedule is timely coordinated with air traffic controllers and airlines using aircraft-to-ground IP ATN data communications of safety-critical air traffic service and business-critical aeronautical operational control messages. Furthermore, unanticipated dynamics of the airspace system are communicated by traffic controllers to aircraft via flight and traffic information services, e.g., real-time weather update, legacy aircraft locations, temporary airspace restrictions, over ADS-B link or IP ATN.

IP networking over broadband satellite and other commercial data links will allow future aircraft to seamlessly exchange digital content with authorized stakeholders at all permitted phases of flight. Use of multihop airborne IP networking further promises to extend coverage for communications between ground systems and aircraft in remote or oceanic regions, both for time-critical ATM information and broadband content.

A. Net-Enabling at Stakeholders

For enhanced operations of future e-enabled aircraft, net-centric capabilities must be enabled for all stakeholders globally [2]. Each stakeholder group may belong to a different organization, geographically located anywhere in the world. These stakeholders together form a global community, interdependent for performance and success of future airspace systems. Securely interconnecting stakeholders through an integrated infrastructure to help each other, without impinging on their independence, goals, and performance needs, however, is a major challenge. For example, the System Wide Information Management (SWIM) infrastructure will enable stakeholders in the United States, such as air traffic management, federal agencies, airlines, weather operations center, to share information (e.g., flight data, weather, airspace surveillance data and infrastructure status), and interoperate with SESAR and other entities worldwide—all together collaborating for coordination of global air traffic [23]. This paper does not further cover net-enabling advances at stakeholders that rely on ground-to-ground networking and its security.

B. Stakeholder Constraints and Their Implications on Aircraft

Solutions for enhancing future aircraft operations must account for aviation stakeholder constraints, including the following.

1) *Highly Regulated Environment*: Civil aviation is subject to a well-established, stringent regulatory environment to assure that aircraft safety and public well being are protected. Regulatory agencies, such as the Federal Aviation Administration (FAA), set mandatory guidelines for aircraft to be considered safe and compliant with minimum performance standards. Any new ground-based or airborne technology for aircraft must be shown to have no adverse impact on safe and expected operation, both under normal conditions and failures. Furthermore, performance standards are emerging to control noise, airport air quality, and atmospheric greenhouse gas emissions.[4] International "green" standards are being established for next several decades, such as reducing air transport CO_2 emissions per year by 1.5% until 2020, capping emissions from 2020 and reducing emissions 50% by 2050 compared to 2005 [14]. Regulations have successfully addressed security risks within e-enabled aircraft certification [9]. Increased use of data links and networks in the future airspace systems may further result in unique considerations of cyber–physical security in future certification and performance standards.
2) *Air Traffic Control Restrictions*: Regulations within ATM systems constrain aircraft movement spatially and temporally, except in free flight and uncontrolled airspace where there is no mandatory control of air traffic. A typical flight includes aircraft parked, taxing, takeoff/landing, or cruising in airspace. Aircraft must comply with tactical (e.g., interaircraft separation) and strategic air traffic rules (e.g., preplanned routes). Based on factors such as location, usage, and flight restrictions, the airspace is divided into segregated classes, e.g., class A to class G in the United States [11]. Flight routes hence are usually predictable with a preflight established trip time, overall making aircraft tend to travel as a group [12].
3) *Aircraft Broadband Connectivity Restrictions*: Regulatory restrictions limit the connectivity between onboard and off-board systems during flight [2], [13]. An airlines operated aircraft may engage in a number of consecutive flight routes, where each flight may require a refresh and/or retrieval of aircraft digital content. During taxing/takeoff/landing however connectivity is limited to critical or essential information communicated over ADS-B and IP ATN data links. During cruise phase, broadband communications over satellite and other data links become available. The most opportune time and location for large volume data transfers is at airport gates, but quick flight turnaround at gates is also critical for airlines.
4) *Stakeholder Interests*: Aviation stakeholder interests may often compete due to conflicting business goals and limited airspace/airport resources, complicating air traffic control decisions. For instance, while an airport's traffic controller may prefer a delay-optimal route, airlines may prefer another fuel-efficient route with more delay. Furthermore, equipment purchase, operation, and maintenance costs are critical factors for financial bottom line of aviation stakeholders. Any new solution must have an attractive cost-benefit ratio and minimize overhead. Further, to obtain a return-of-investment on existing technology and processes, new solutions must leverage these and enable airspace system modernization [1].

[4]http://www.boeing.com/commercial/noise

C. Threat Model and Assumptions

In this airspace system, an adversary may attempt to lower performance and cause economic damage by creating unwarranted safety concerns that cause flight delays and passenger anxiety. The adversary is capable of external cyber–physical exploits, such as active manipulation of data, node impersonation, radio-frequency jamming, and passive eavesdropping, and internal (insider) exploits such as compromising cyber elements. Credible examples of potential misuse by such an adversary in future aircraft include: malware to infect an aircraft system, exploit of onboard wireless for unauthorized access to aircraft system interfaces, intentional denial of service of wireless interfaces to aircraft systems, intentional denial of service of safety critical systems, misuse of personal devices that access aircraft systems, and misuse of off-board data links to access or disrupt aircraft system interfaces.

For simplifying exposition in this paper, cyber–physical threats are limited to be over aircraft data links and networks. Insider threat to future aircraft can be very effectively deterred by enforcing legal regulations, sufficiently safeguarded against with physical, logical, and organizational inhibitors, checks, and control. Nevertheless, consideration of the insider threat can enhance the depth and completeness of cyber–physical security analysis as well as level of protection.

Processes at each entity in the airspace system are operating as designed and expected, and the entire system is administered appropriately, e.g., proper assignment and management of access privileges at each entity, proper management and protection of passwords, cryptographic and security quantities. Aircraft operators are assumed to

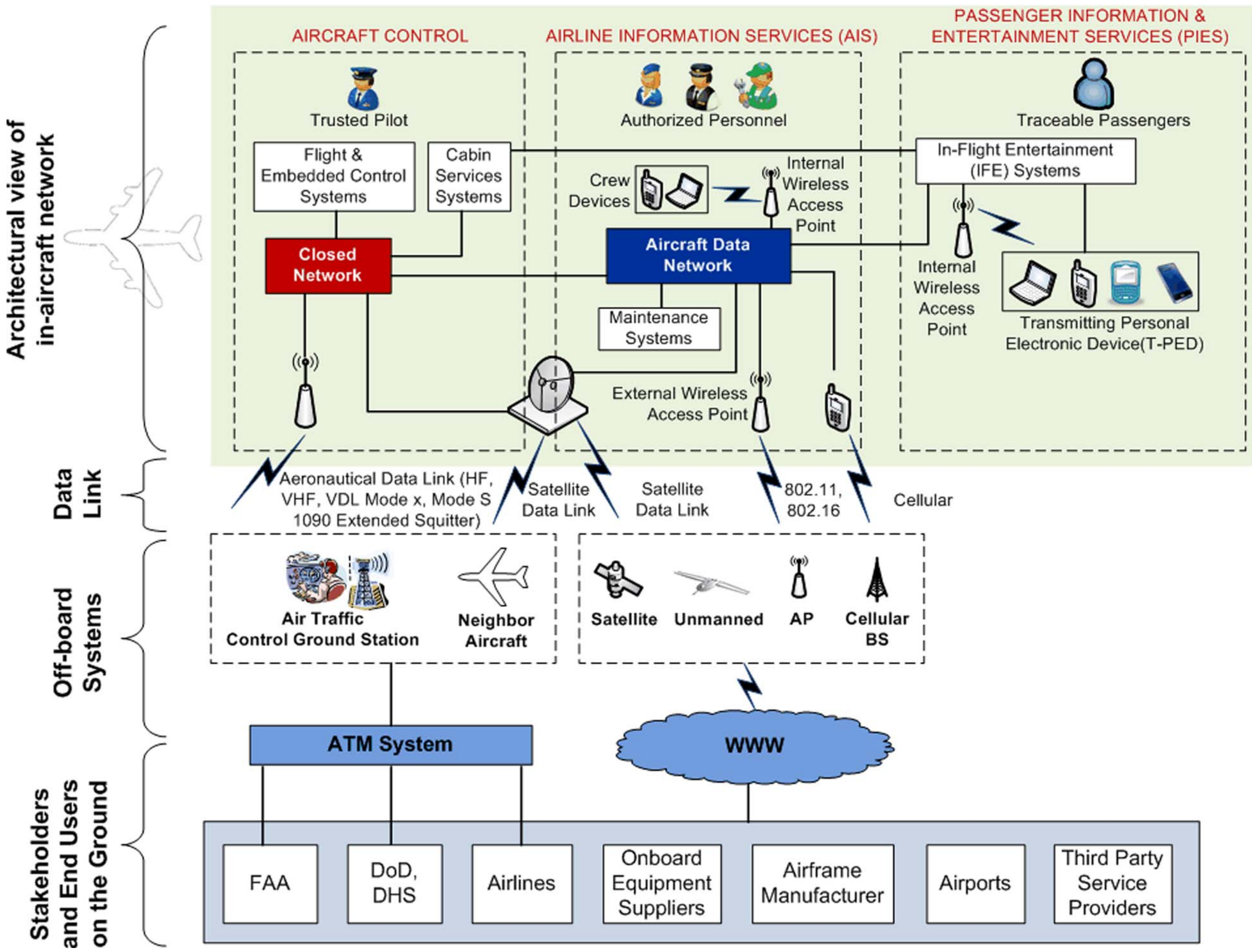

Fig. 3. ***Illustration of an abstract connectivity architecture for the future e-enabled aircraft in commercial aviation.***

operate correctly, reliably managing software configuration and other digital content of their fleet. Air traffic controllers and flight deck crew are trusted with various ATM management tasks.

In the presence of such an adversary model and high-level system assumptions, it is vital to protect the integrity, authenticity, and availability of aircraft data communications. Passengers and some airspace users may also be privacy sensitive, requiring confidentiality and anonymity for communications including personally identifiable information.

III. FUTURE E-ENABLED AIRCRAFT NETWORK

A. Integrating Across In-Aircraft Network Domains

All systems in the aircraft, generally, can be categorized based on their certification level into the following broad areas: ATM, flight operations, airline administrative, and passenger services [2]. Onboard systems, in the past, were physically separated so that security threat vectors were from outside supporting systems. However, going forward in the integrated systems environment, it is imperative that these multiple areas form a tightly integrated networked environment without compromising their integrity.

The top part of Fig. 3 illustrates high-level onboard network architecture for the future e-enabled aircraft, with three logical domains: aircraft control, airline information services (AIS), and passenger information and entertainment services (PIES) [2], [15], [16]. Related stakeholders are also shown. Nodes are arranged in multiple, independent networks, based on their functions. The aircraft control domain includes avionics and control systems at all levels of criticality—systems both for safety-critical operations and those whose failure or malfunction may impact comfort and efficiency (e.g., cabin lights). AIS systems support business operations, an area which is growing on commercial aircraft as airlines find it increasingly cost effective to carry equipment providing automated support to cabin and flight crews. PIES includes in-flight entertainment systems and "brought in" passenger devices, e.g., laptops, cell phones, medical devices, RFID tags, that do not support or impact any flight function [39]. Aircraft maintenance systems onboard require special consideration, not addressed here, since maintenance support is required for resources in every domain. A major challenge is that while it is undesirable to create multiple isolated maintenance systems, sufficient connection of all networks solely for maintenance can increase potential threats.

All cross-domain communications are secured at multiple layers using sufficient physical, logical, organizational inhibitors, e.g., network firewalls, routers, switches, monitors, and usage policies for wireless devices carried onboard [15]–[17]. These measures along with other host-based mechanisms (e.g., filtering, redundant storage, tamper-proof logging of security-relevant actions) ensure the flight control domain is operational and closed to passenger domain as well as protected against unauthorized access and eavesdropping from cabin domain [18]. Similarly, cabin domain is secured from passenger domain and any potential unintentional interference from passenger devices is adequately mitigated [19].

B. Off-Board Communications of the In-Aircraft Network

For regulatory reasons and continuity/longevity of onboard avionics, the onboard networks are required to have different treatments of their aircraft-to-ground traffic and must use distinct data links and service providers. As shown in Fig. 3, communications with air traffic controllers, neighbor equipped aircraft, and other stakeholders in the ATM system is performed via aeronautical protocols over different RF data links, i.e., satellite relay (UHF), terrestrial [HF, VHF Data Link (VDL) Mode 2/3/4, IEEE 802.16e]. ADS-B offers a Mode-S 1090-MHz Extended Squitter (1090-ES) and 978-MHz Universal Access Transceiver (UAT) data links which can achieve narrow bandwidth communication beyond 100 miles [4], [20]. The flight deck communications, i.e., air traffic services (ATS) and airline operational communications (AOC) [21], include voice/data exchanges for air traffic control, weather and map information, in-flight security, and other services related to flight safety, regularity, and efficiency. The aircraft will possess a globally unique, permanent digital identity, such as the 24-b International Civil Aviation Organization (ICAO) address, tail number, flight number [20], included in flight deck communications from aircraft for security in controlled airspaces.

AIS off-board communications, i.e., airline administrative communications (AAC) [21], includes business-critical voice and data exchanges for flight logistics, aircraft maintenance, etc. For these communications, aircraft is connected to the Internet, over satellite or unmanned system relays during en-route, and emerging IEEE 802 wireless (e.g., 802.11, 802.16) or cellular networks at airports. Cellular network connectivity is today available worldwide, but network bandwidth may vary up to 500 kb/s based on second to fourth generation cellular technology (i.e., 2G–4G). The narrow bandwidth connectivity can support only some limited functionality for future aircraft. For greater functionality at airports, low-cost, high bandwidth connectivity (over 100 Mb/s) is needed, so that operational, maintenance, and in-flight entertainment data can be reliably transferred at the gate within the allotted flight turnaround time. WiFi and WiMax are being implemented at some airports, but both of these IEEE 802 standards are mired in global spectrum issues impeding their applicability as a universal solution. Physical connectivity to the aircraft parked at gate, using gigabit copper, power line, or fiber cables is being explored, but has been limited in success due to maintainability and logistical issues. Therefore, future aircraft needs broadband, reliable, secure connectivity on the ground to the Internet that can be inexpensively deployed worldwide.

Connectivity for airline passenger communications (APC) [21] will grow substantially in the future, with passengers demanding high bandwidth access to data and other amenities they are accustomed to in their home/office environment. To enable such connectivity, passenger devices must securely access third party services using broadband IP network links available at high altitudes and remote regions [2].

C. Future Directions for the In-Aircraft Network

Weight reduction will continue to be one of the highest priority "green" initiatives to reduce fuel burn and emissions rates in future e-enabled aircraft design. Towards this objective, apart from lightweight airframe design (e.g., composite structures instead of metal), an integrated modular architecture with high-end processors is employed to reduce the number of embedded systems, power consumption, and amount of cabling in an aircraft [22].

To lower system size, weight, cost, and volume, standards and guidelines are being developed to integrate the three logical domains of the in-aircraft network so that they securely share data links [2]. It is highly likely that flight-critical data will always be separated from other types of information flowing through aircraft. Bandwidth-intensive flight-essential applications, however, may share a common broadband link with airline domain, e.g., transmission of 3-D weather maps and cabin surveillance video. Furthermore, upon potential link failures, these applications may fallback to an available commercial broadband data link of the airline domain while non-flight-essential applications may use a lower latency link.

Advances in wireless technology and energy harvesting are also anticipated in some onboard systems to eliminate network/power wiring and system weight, if these advances are introduced safely and efficiently [7]. A wireless-sensor-based solution, for instance, must meet the following key criteria: 1) frequency spectrum

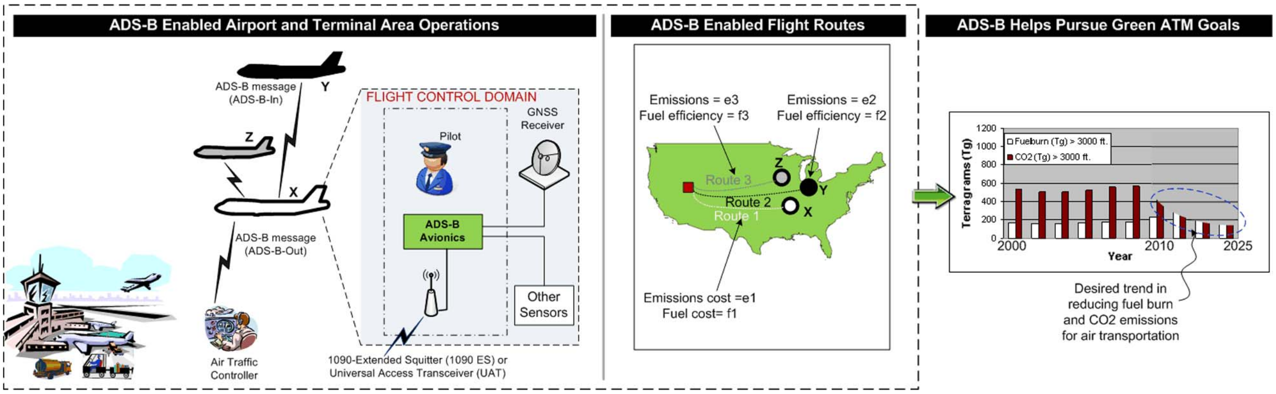

Fig. 4. ***Illustration of ADS-B modes and benefits for future ATM systems.***

compliance around the world; 2) extremely low energy or energy harvested so that no power wires are required; and 3) demonstrate high reliability, availability, and security. One approach towards meeting these criteria is to dedicate some spectrum for aviation sensor applications [2]. Increasing the use of wireless communications onboard, including for critical functions, is also being investigated, e.g., the fly-by-wireless program [7]. Furthermore, applications for aircraft digital content delivery are negating weight of onboard storage media [6].

The remainder of this paper focuses on challenging issues and risks in using off-board data communications and networking of e-enabled aircraft for managing air traffic and aircraft digital content in future airspace systems. Under the assumption of closed, secure backbone data networks for onboard systems, only the different data links and networking with off-board systems are considered most susceptible to cyber–physical threats.

IV. CHALLENGES IN ADS-B FOR AIRCRAFT MOBILITY

ADS-B infrastructure is today being deployed across the world. It brings a paradigm shift from radar to data-based air traffic surveillance, transforming the way aircraft are monitored for air traffic control in the next-generation ATM systems [4]. The basic idea of ADS-B is that an aircraft computes its present 3-D position using GPS and uses a data link to broadcast a position message containing the computed position, time, and spatial data such as velocity and intent. The message includes a digital identifier of aircraft, e.g., 24-b ICAO address. The resulting frequent, accurate, and widely available position feedback from aircraft can, among other benefits, reduce interaircraft spacing to less than 3 nmi (nautical mile). Such a spacing reduction translates to airspace system performance gains in terms of capacity and efficiency.

As shown in Fig. 4, *ADS-B-Out* makes an aircraft periodically broadcast position messages and air traffic controllers on the ground can use these messages to monitor airspace, i.e., *ground surveillance*. A more revolutionary offering is *ADS-B-In*-based *airborne surveillance* capability which makes an aircraft use position messages to monitor neighbors in airspace and airports, improving pilot situational awareness and alertness. In the longer term, ADS-B-In is envisioned to help autonomous spacing of aircraft. Future ATM systems rely on the accuracy and performance of ADS-B surveillance, for example, fuel and time efficient operations at/near high density airports, en-route, and remote airspace.

Potential "cyber–physical" security vulnerabilities in both GPS and ADS-B technology, however, remain to be assessed and resolved, impeding rapid design, deployment, and beneficial usage of many ADS-B applications [24], [30], [37].

A. Perceived Threats to ADS-B

Vulnerabilities in ADS-B applications emerge because of the dependency of critical ATM tasks on the integrity of ADS-B data delivered over a shared data link. To create unwarranted flight delays, safety concerns, and operational costs, adversaries may attempt to intentionally engineer errors in surveillance by disrupting GPS or by corrupting and spoofing ADS-B data, for example, create misleading flight route conflicts by broadcasting false ADS-B positions. Such cyber-induced failure conditions may cause the ATM system to degrade in surveillance accuracy and/or performance gain.

Most safety threatening situations in surveillance, such as midair collisions, can be mitigated by relying on onboard navigational systems, such as traffic collision avoidance system (TCAS) transceiver which interrogates corresponding transceivers of neighbor aircraft [25]. On the other hand, secure ADS-B surveillance data can serve as an additional protection layer for TCAS [25]. For example, flight plan intent provided by ADS-B can benefit conflict detection and collision avoidance algorithms.

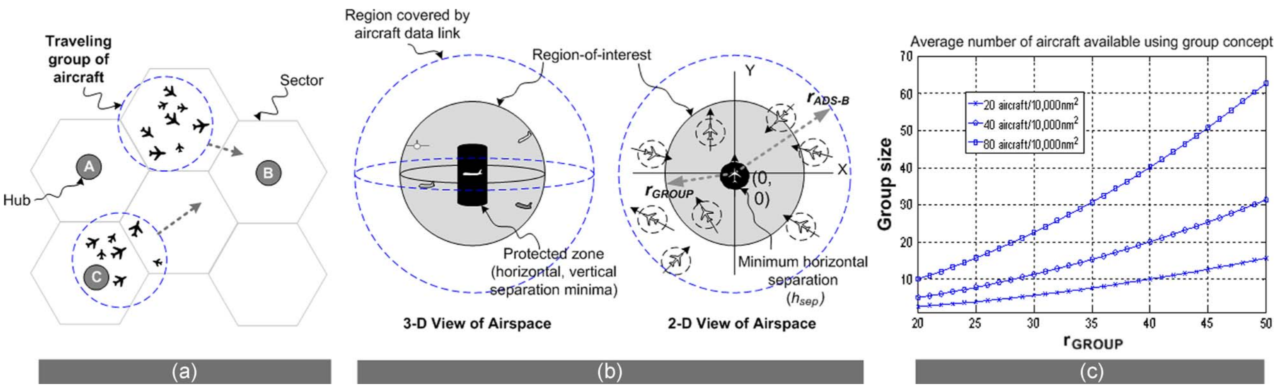

Fig. 5. *Group concept for validating ADS-B-In messages and preventing unauthorized location tracking of ADS-B equipped aircraft.*

B. Current State of GPS Assurance for ADS-B

GPS will in the future serve as a primary information source for aircraft navigation. Accuracy, integrity, availability, and continuity of service of GPS, hence, must be ensured in the presence of both natural and malicious disruptions. A high-level assessment of potential impact of GPS vulnerabilities, such as intentional jamming interference, on aviation safety is in [27]. Further, accuracy of GPS readings can be achieved by making aircraft include spatial precision parameters in an ADS-B-Out message to establish error margins for ADS-B-position-based computations at the receivers [26]. Furthermore, unintentional GPS degradations around airports can be compensated using the ground-based augmentation system (GBAS) which operates in the very-high-frequency (VHF) range and transmits differential corrections in GPS receiver measurements to aircraft approaching or leaving the airport, thereby preserving or enhancing accuracy, integrity, instantaneous availability, and continuity of service of GPS at that airport [27].

Detectable failure or outages of GPS can be addressed by employing secure land-based alternatives as a backup, such as terrestrial radio navigation systems operating in low-frequency range (e.g., 100 KHz) [27], [38]. Primary radar technology is expected to remain operational for military, national security, and law enforcement missions. Hence, primary and secondary surveillance radar may also serve as land-based alternatives for sourcing aircraft position information, either to help air traffic surveillance during potential failures of GPS for validation of ADS-B-Out data in radar-covered regions. Such an approach, however, requires ability to fuse data from radar and ADS-B.

Upon detection of a GPS outage or disruption, a strategy is needed to gracefully degrade aircraft navigation to a less accurate coverage of the backup system. As discussed in [28], the interaircraft spacing must be gradually increased over a short time period to ensure safe transition to the degraded surveillance accuracy. Furthermore, recovery strategies are needed for outages in nonradar covered regions.

C. Current State of ADS-B Security

Recently, cyber security assessment of ADS-B-based ground surveillance is in FAA final rule for ADS-B-Out mode [29]. It is evident that primary and secondary surveillance radar infrastructures will serve to verify the integrity and smoothly compensate for the loss of ADS-B-Out data in terminal areas and en route. Additionally, multilateration offers a potential alternative where at least four ground stations can jointly verify the 3-D position in a received ADS-B message by estimating the position from the received signal properties (e.g., signal strength, time of arrival). A simple solution approach is to combine two multilaterations at the ground stations, one from use of time of arrival of the ADS-B signal and another from ADS-B position data. For regions with less than four ground stations, a promising approach for aircraft position verification is to use ADS-B enabled multilateration in combination with Kalman filter estimation of the flight trajectory based on the bearing information of the source making the position claim [31], [33]. This approach however relies on a dedicated omnidirectional antenna in the aircraft for deriving source heading.

Cryptography can protect ADS-B communications between aircraft and air traffic controllers. Use of symmetric-key-based solutions offers an efficient way to protect integrity, authenticity, and confidentiality of ADS-B-Out messages from aircraft as well as flight or traffic information from ground controllers. One efficient approach is to use a keyed hash or message authentication code [32]. However, symmetric-key-based solutions require aircraft and air traffic controllers to preshare a secret. A potential alternative is to use digital signatures and public key encryption. Efficient asymmetric cryptography, e.g., elliptic curve cryptography, solutions can protect traffic and flight information sent from traffic controller to aircraft over ADS-B data link, since allowed message sizes is in order of several hundred bytes (e.g., 556 B for UAT). However, use of these solutions is challenging for protection of ADS-B-Out data from aircraft, because of smaller payload (e.g., 34 B for UAT). Furthermore, solutions require availability and management of a public key infrastructure, which may pose scalability, cost-benefit challenges for ATM systems [6].

ADS-B-In-based airborne surveillance however is impeded by the current lack of solutions for verifying position data in an ADS-B message received at aircraft. Addressing this gap can help improve ADS-B benefits and applications.

D. Mitigation of Threats to ADS-B-In Applications

In [24], a solution to validate ADS-B-In message integrity is proposed based on the existence of traveling groups of aircraft. As seen in Fig. 5(a), aircraft mobility is spatially/temporally restricted and spatially dependent. These constraints result in the group navigation property, i.e., aircraft in geographical proximity navigate as a group for a time period. Aircraft in a navigating group will have the same average velocity and similar direction. Furthermore, a group of navigating aircraft within distance $0.5r_{\text{ADS-B}}$, where $r_{\text{ADS-B}}$ is ADS-B data link range: 1) are geographically proximate, 2) maintain full mobile IP network connectivity, and 3) do not suffer any significant signal degradation in group communications [12]; see Fig. 5(b).

Based on the above observations, the group concept is proposed where neighboring aircraft form a reliable communications group during navigation. Each group member must be able to hear broadcast of every other member so that a fully connected IP network graph exists within the group. Since aircraft in a group will move relative to each other and on average have the same velocity, a group can be represented by a single aircraft, i.e., group leader. The full IP connectivity within the group allows group communications for members to establish shared secrets and perform joint operations such as validating ADS-B-In messages. Fig. 5(c) shows feasibility of the group concept.

When an ADS-B signal from an aircraft P is received by four or more aircraft, $\{V\} = \{V1, V2, V3, V4\}$, then $\{V\}$ can use time-difference-of-arrival technique to estimate the 3-D position of P. In this technique, $\{V\}$ must share the times they received the signal from P, compute the differences between these times, and then perform multilateration computations to estimate actual position of P. An underlying principle of this technique is that RF signals cannot be sent faster than the speed of light c. Therefore, if four or more verifiers can communicate the time instances at which they received ADS-B signal of P, they can independently estimate P's actual position and verify claimed position in ADS-B message of P.

Together, the group concept and time-difference-of-arrival-based multilateration can help an aircraft verify that the position in a received ADS-B-In message is valid. A group of geographically proximate aircraft can consecutively receive an ADS-B-In message from a prover. The group of aircraft can then use aircraft-to-aircraft IP network links to securely communicate the times of arrival of a received ADS-B signal, use the communicated measurements to self-estimate actual position of the source of the ADS-B signal, verify if the ADS-B position data matches the estimated position, and if needed perform secure voting to invalidate any potentially corrupt or spoofed ADS-B-In message. The group concept also serves to protect against misuse of ADS-B as described next.

E. Preventing Misuse of ADS-B Data

Data in ADS-B messages can be misused by unauthorized entities from outside the airspace system to identify, locate, and track target aircraft. Such unauthorized location tracking raises potential privacy concerns for some airspace users, e.g., general aviation aircraft operators [29]. Location trajectories of a private aircraft, when correlated with public databases and geographic maps, can help in accurately inferring business or political developments, travel intent, and profiling of users personal location preferences and interests. Such tracking may also expose high-value aircraft, e.g., law enforcement, to physical threats from ground.

The "privacy mode" option available for the ADS-B UAT data link can allow private aircraft to operate anonymously when operating under visual flight rules mode in uncontrolled airspace such as class G or class A [20], [29]. The use of ADS-B in uncontrolled airspaces is assumed to be for situational awareness enhancement and not for safety-critical applications such as assurance of interaircraft spacing.

The privacy mode generates a 24-b pseudonym for the aircraft, such as by making the aircraft compute a random identifier as a *pseudonym*. A practical solution is outlined as a standard in [20], using which the aircraft computes the pseudonym as a function of a random quantity, e.g., the location and/or time of use of pseudonym, and the 24-b ICAO identifier. Because air traffic controllers know the ICAO address of the aircraft as well as can record and verify ADS-B broadcasts from the aircraft, they can maintain airspace security when emergency events occur. Another promising approach is allow aircraft to share an encryption key with air traffic control [32], and use this key to secretly communicate the random quantity used for the pseudonym.

The use of pseudonyms however cannot provide location untraceability when there is a strong spatial and temporal correlation between aircraft locations due to underlying predictable mobility of aircraft and the short intermessage period of ADS-B (e.g., 0.5 s) [12]. The strong correlation between consecutive positions may be leveraged to associate pseudonyms with an observed permanent identifier and threaten the aircraft operator's location privacy.

A promising solution for mitigating location tracking is to use a random silent period and update the pseudonym [12]. However, this solution enlarges the ADS-B broadcast period, i.e., reduces timely availability of ADS-B messages. In order to achieve a random time period for pseudonym update without degrading ADS-B performance, a solution approach is to leverage the group concept described earlier.

The group of aircraft can continue to broadcast ADS-B-Out messages with pseudonyms, while coordinating to be represented by a common valid group identifier for IP communications as well as establishing secrets on-the-fly for group-based operations. Each aircraft in a group reduces its transmission range for ADS-B and IP network links to reach only other members of the group (such as 6–10 nmi). Each group has a leader with a larger transmission range that is sufficient to reach airborne and ground nodes (such as 100 nmi). A candidate for group leader is an aircraft that does not require location privacy. In such a solution, the adversary can at best know only the group's identifier and the group leader's location. Given that the group identifier is only traceable to a navigating group of aircraft and that members can self-update their pseudonyms while participating in the group, each member can potentially achieve an extended random time period for a pseudonym update. This random period is equal to the time duration for which the member is in the group. The traffic controllers can still identify and accurately trace valid nodes in the sky as a cluster, while eavesdroppers can at best randomly guess the trajectories of the airborne nodes.

F. Open Challenges in Securing ADS-B

"Intelligent jamming" strategies present potential risks to ADS-B availability and their impact of ATM system performance must be studied. For example, selective intentional jamming may disrupt specific air traffic flows or sectors in airspace systems [43].

ADS-B-Out is today viewed as an added equipment for air traffic control tasks and not as a replacement for existing aircraft transponders. An understanding of how the ATM system can mitigate, detect, and respond to ADS-B vulnerability exploits can streamline the effort to bring lesser dependence on legacy aircraft transponders and ground radar infrastructure in the next two decades and beyond.

Upon detection of ADS-B failures, solutions must ensure performance and accuracy of surveillance gracefully degrade to a backup system, both in radar and nonradar covered airspaces. The choice of inter-aircraft spacing which determines the degradation in surveillance accuracy and the required transition time, must account for future air traffic densities. For instance, degrading to secondary surveillance radar is not efficient and effective for all detections of cyber–physical exploits, because the pessimistic spacing of secondary radar can allow threats to enlarge impact on the system, e.g., from a terminal area to an entire traffic control center. The proposed tool helps the design of early, effective response to detected ADS-B exploits that ensures graceful degradation of ATM capacity and efficiency with increase in actual impact of exploits. Additionally, metrics and models are needed to accurately measure and predict performance degradations from ADS-B disruptions.

V. CHALLENGES IN IP ATN FOR AIRCRAFT

IP ATN can bring a paradigm shift from a voice-based to a data-based management of air traffic in the next two decades. An emerging vision of the IP ATN is that IPv6-standards-based networks will be used for ground-to-ground, aircraft-to-ground, and aircraft-to-aircraft data communications [2], [34]–[36]. IPv6 advances promise widespread commercial product and network support for the successful worldwide deployment and use of new, high data rate ATN applications.

A. Current State of ATN Data Links

Existing ATN data links already occupy the allocated communications spectrum at or near capacity. Additionally, these data links are shared with unmanned aircraft operators and maritime applications. Consequently, most of the existing data links have very low throughput of 1–30 kb/s. Newly available data links such as satellite Ku/Ka-bands in the 1–50 Mb and 3G & 4G air-to-ground links in the 1–20-Mb range are in service today; older cellular link services in the 0.1–5 Mb, new terrestrial and satellite relay links in the gigabit ranges will be deployable within the next decade [35].

New aircraft-to-ground data link standards are emerging for IP ATN. Aeronautical Radio Inc. (ARINC) is defining exchanges necessary to authenticate and encrypt IP-based aircraft-to-ground links as well as the messaging structure for application-to-application message passing, critical to ensure compatibility with existing message-based ATN links [41]. Required communication performance (RCP) standard is defining the certified use of these new data links in the ATN [42]. The first RCP implementation guidance presents a complex process that allocates performance requirements to each portion of the end-to-end path. However, application guidance on how, when, and where to use RCP and how to enable utility in actual operations, is still lacking. While RCP can help in certifying ATN data links in the future, it is not clear how to obtain operational authorization for applications.

B. Emerging Challenges in IP Networking of Aircraft

1) Network Routing: Current Internet is not designed to support a large volume of mobile platforms moving globally at 600 miles/h, due to the potential routing disruptions it causes on the Internet. Boeing is currently leading an effort through the IETF to rectify this problem by developing a new standard based on IPv6 and mobile IP technologies. ICAO has identified the network mobility (NEMO) technology as a promising candidate for use in supporting global, IP-based mobile aircraft networking [34]. Major unresolved challenges remain in NEMO, such as the lack of route optimization support and potentially complex configuration requirements in a large airport environment with multiple service providers and 25 or more airlines sharing the same infrastructure. Furthermore, multihop routing of messages between aircraft is a new concept for civil aviation that remains to be investigated for improving coverage of IP ATN.

2) Aircraft DNS Naming Standard: Currently aircraft systems have preassigned fixed IP addresses for their devices, which may not be a good way to go forward. A more promising approach would be to emulate a corporate environment with domain naming system and dynamic host configuration protocol servers that can dynamically assign IP addresses. Such an approach can help airlines with a consistent operational scenario across their fleet with domain names instead of fixed IP addresses. Standards activities, such as ARINC Network Infrastructure and Security subcommittee [41], are helping to address these future issues.

3) Standard Aircraft Network Identity: Today airframe manufacturers and airlines identify their aircraft in many different ways. Going forward the airlines should have a common identity format, process, and other key aircraft identity related parameters so that systems and tools do not have to be customized in each case. These identities should be electronically embedded in the aircraft so that the aircraft can be uniquely identified at all times reliably, enabling transparent handover across subnetworks for seamless mobility. Furthermore, the identities should link directly to the unique ICAO aircraft transponder code critical in ATM. However, if it is embedded in hardware then there is a risk that it may not be properly restored when the part is replaced without a robust process in place. The Air Transport Association Digital Security Working Group is one of major efforts working to resolve these issues.

4) IP Network Addressing: A major concern at present is that there is no large IPv6 network infrastructure deployment anywhere in the world Additionally, 128-b IPv6 addresses are not human friendly (e.g., 2001:db8:85a3::8a2e:370:7334) [35]. It consists of 16, 8-b fields coded in hexadecimal, with first 64 b as the network portion and last 64 b the unit portion. The unit and network parts of the address are fixed at these 64-b boundaries and are unchangeable in IPv6 unlike IPv4. To manage the complexity of network addressing, one approach is to exclusively design Internet names for aircraft[5] so that these names can contain aircraft transponder code and be translated to an IP address [35]. Furthermore, it is important that naming be standardized globally for seamless operations (e.g., flxx.airlineyy.atn), and that ATN is assigned a global IPv6 address block to keep infrastructure and operational costs low. A fragmented ATN address space imposes potential risks on airborne IP networks.

5) IPv4 and IPv6 Support Worldwide: Although some countries are aggressively pursuing the conversion from IPv4 to IPv6, it is a given that around the world some countries may still be in IPv4 for the next 10–20 years. IPv4 applications cannot directly interoperate with IPv6 applications, however, IPv4 and IPv6 networks do coexist on the same equipment and media (wires, links). It is imperative therefore that future aircraft will have to seamlessly connect and operate in countries with IPv4 as well as IPv6 infrastructure. Standards are needed to maintain this compatibility globally.

6) Cognitive Connectivity: Technologies are needed for aircraft at any location to connect to any specific link dynamically based on its present security level, reliability, bandwidth, and other characteristics, and transmit the right type of data appropriate to that location globally.

C. IP ATN Security Issues

Recently, the ICAO has recognized the need for cyber security considerations in IP ATN, and recommended that ATN data be protected from unauthorized disclosure, modification, or deletion and ATN resources be protected from unauthorized use and denial of service. Achieving these security objectives is a major task due to the mixed criticality of data transmitted over IP ATN. Furthermore, security is challenged in a global heterogeneous system environment due to complex issues such as dependency on a global public key infrastructure (PKI) "czar," lack of interoperability of IPSec and PKI at vendors, and lack of secure digital identity associated with messages without PKI [35].

The foundation for IP ATN security is a "trusted identity." Several transponder communications with air traffic controllers today are based on an "honor system" which assumes implicit authentication of the communicating parties. With increased use of IP ATN in future ATM systems, however, an authentic, unforgable identity is essential as a building block for secure communications [35]. Several open problems emerge, such as using transponder code in the aircraft PKI certificate as a trusted identity, embedding transponder code in IPv6 address to allow aircraft's network address to be the trusted identity.

[5]www.airframerxx.aero

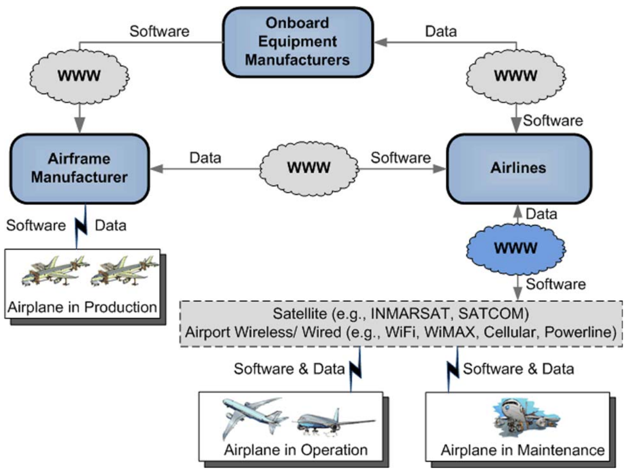

Fig. 6. ***End-to-end distribution of aircraft software and maintenance data.***

VI. CHALLENGES IN AIRCRAFT DIGITAL CONTENT DELIVERY

IP networking of aircraft with off-board systems also promises to enable digital content delivery that enhances seamless aircraft operations in future net-centric airspace systems. Due to the type and role of digital content in future aircraft, the content can be expected to be increasingly susceptible to threats. For instance, consider safety-assured software developed by onboard equipment manufacturers using guidelines in RTCA DO-178B. This includes safety-critical software (e.g., flight control computer) and business-critical software (e.g., cabin light, in-flight entertainment systems). Malfunction of such software can degrade aircraft performance and/or create unwarranted flight delays/costs as well as visible disruptions in the operation of onboard systems (e.g., cabin light flickering). Further, some aircraft digital content may be corrupted to cause economic loss from false alarms or late fault detections. Some content may also include intellectual property (e.g., RFID data, in-flight movies, passenger health data). Aircraft digital content hence must be protected during distribution over IP networks.

A. Electronic Distribution of Aircraft Digital Content

Fig. 6 shows complexity of a system handling future aircraft digital content integrated across stakeholder domains. The end-to-end system must meet security requirements against perceived threats while including network infrastructures, personnel, and business processes of each domain. The system must also consistently exhibit some security properties at suppliers, manufacturer, airlines, airports, and aircraft. For software distribution, onboard equipment manufacturers initiate the distribution and aircraft finally receives the software over an IP network link. Delivered software is uploaded into onboard systems via a process protected by logical and physical controls, e.g., loading is performed only during maintenance mode by authorized personnel using authorized equipment. Additional checks may detect corrupted or unwanted software: cyclic redundancy code error check by embedded systems to detect accidental software modifications, onboard generated configuration report check to verify distributed software matches intended aircraft configuration, and compatibility check of loaded software with destination hardware/software.

Digital signatures offer a solution mechanism to protect the end-to-end integrity and authenticity of aircraft digital content in the presence of threats [6]. For example, threats from software incompatibility across versions and aircraft models can be mitigated by including metadata with versions and timestamps to reject outdated software and including intended destinations to reject diverted software. ARINC is currently defining standards and industrial implementations and deployments are ongoing, such as at Boeing and Airbus [6].

B. Open Challenges in Digital Content Delivery

Formal methods, e.g., model checking, can help establish with high confidence the immunity of this system to threats. Formally assessing end-to-end security however is complex, time consuming, and expensive, requiring high assurance of a large number of connected components. A promising solution is to transport software from source to destination via interaction of instances of a single reusable certified core component communicating over insecure links [6].

Further, a key concern is broadband off-board connectivity to IP networks. Today there is a shortage of dedicated bandwidth for aviation networks resulting in the use of multiple data links and antennas on the aircraft. While the growing availability, flexibility, and cost effectiveness of commercial data links (e.g., IEEE 802.11, IEEE 802.16, cellular) offer promising alternatives, using uncertified data links for certified onboard systems is a major challenge. Additionally, digital content delivery requires future aircraft to interact and exchange information with a wide community of stakeholders who have off-board systems that are geographically dispersed and heterogeneous in capability. Integrating the future aircraft with such a large-scale, net-centric system of systems is a significant issue.

VII. AIRCRAFT CERTIFICATION AND STANDARDS

Apart from the enormous technical challenges ahead for tightening integration within aircraft and between aircraft and off-board systems, there are equally challenging regulatory aspects. To maximize future airspace system performance, technical and regulatory advances must be simultaneously made to bridge a perceived growing disconnect.

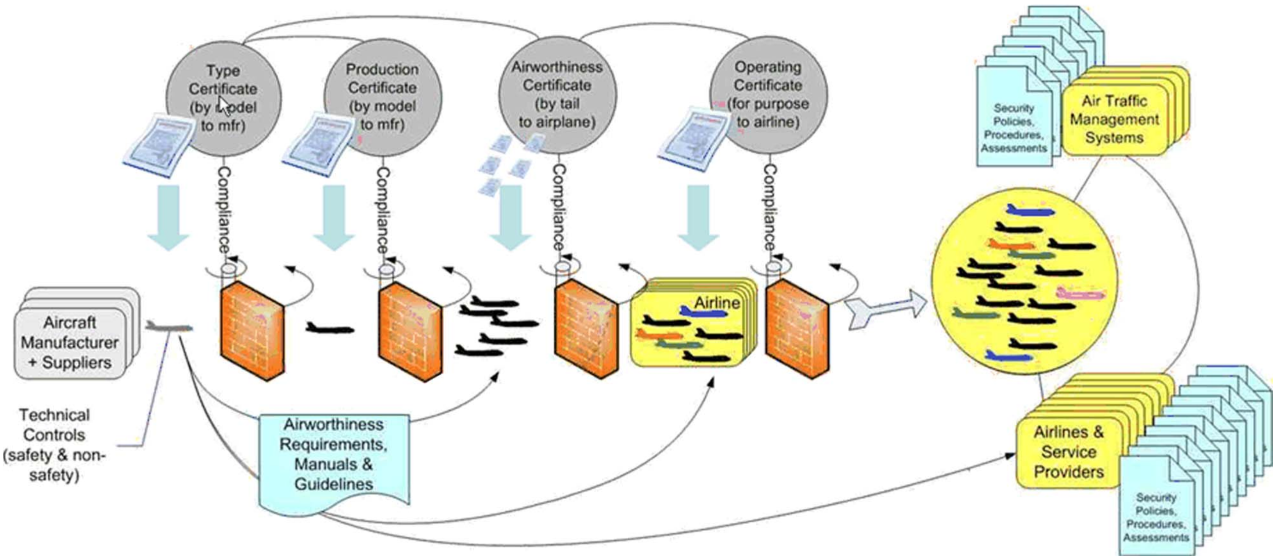

Fig. 7. *Illustration of a typical commercial aircraft certification and deployment process.*

A. Growing Disconnect Between Technology and Regulation

Many of the existing minimum performance standards and regulations for assuring safe flight (e.g., interaircraft spacing) may be degraded by new operational benefits envisioned in future airspace systems (e.g., substantially reduced interaircraft spacing with ADS-B). Further, off-board systems potentially impacting how safely aircraft operate will heavily use commercial grade hardware and software which cannot be assured using the existing standards for systems responsible for safety. Consequently, a fundamental challenge for the next two decades and beyond is updating regulations and standards to meet new safety objectives, development assurance, and performance demands of future aircraft and airspace systems.

B. Typical Aircraft Certification Process

Aircraft certification is the well-defined process whereby an applicant requests an approval from an aviation regulatory authority, such as the FAA or the European Aviation Safety Agency (EASA), for manufacturing a new aircraft model or making changes to an existing aircraft either as supplements or amendments. Aircraft certification processes use approved standards, guidance, tests, methods, procedures as well as data submittals and plan documentation to achieve regulatory approval for aircraft type certification.[6]

However, as illustrated in Fig. 7, the process of rolling out an aircraft into service involves far more steps than receiving a type certificate. For each step, a certificate is granted by a regulator after an applicant demonstrates that they meet the requirements which, with the addition of cyber security concerns, inevitably will require that regulatory requirements unique to the regulatory oversight role for each certificate be developed and compliance shown. Add to that the implications of hundreds of ATM systems, airline information technology and operational systems, and service provider infrastructure, and the simple notion of "end-to-end" security is revealed to be a tantalizing concept, but an overwhelming task. To be effective the specific allocation of each portion of this task must be developed to fit the security requirements of the roles and responsibilities for each segment, along with the upstream "provider" and downstream "consumer" hand-offs. This is a major challenge which requires significant coordination and consultation between security and aeronautical specialists. However, it is a necessary "modularization" of the problem for faster progress through parallel effort, better results through ownership of security as part of each segment, and less disruption because the changes are, wherever possible, made within existing regulatory and industry frameworks rather than across them.

Airworthiness safety is considered and evaluated under the concept of an aircraft being airworthy [40]. The two conditions necessary for issuance of an airworthiness certificate are: 1) aircraft must conform to its type design, and 2) aircraft must be in a condition for safe operation. Conformity to type design is attained when the aircraft is consistent with data of the type certificate. Whether an aircraft is in a safe operating condition depends on 1) condition of the aircraft relative to wear and deterioration, e.g., issues of skin corrosion and tire wear, and 2) condition of complex digital systems, including software and hardware, relative to their ability to properly operate. If one or more of these conditions are not met, the aircraft is not considered airworthy.

[6]http://www.faa.gov

Airworthiness safety of future aircraft with cyber elements must be certifiable in the presence of potential for malicious disruption and misuse. Physical security threats potentially impede condition of the aircraft relative to wear and tear, while cyber exploits may impair the condition of the digital systems to operate properly. Thus cyber–physical threats potentially impede the aircraft's condition to operate safely, warranting the need for airworthiness security considerations within aircraft development and certification process.

C. Security Considerations in Future Aircraft Certification

Airworthiness security can be defined as the protection of the airworthiness of an aircraft against cyber–physical threats from harmful human action (accidental, casual, or intentional) using access, use, disclosure, disruption, modification, or destruction of data and/or data interfaces of the aircraft [9]. This includes consequences of malware, maliciously corrupted data, unauthorized access of other systems to aircraft systems, and denial of service of aircraft networks and computers.

Current guidance for aircraft certification must be augmented to handle cyber–physical threats to airworthiness and flight safety of future aircraft. Such guidance must be developed in the context of existing regulations, such as Title 14 Code of Federal Regulations Part 25 "Transport Category Aircraft." The guidance must consider threats that emerge throughout the lifecycle of a commercial aircraft type, i.e., project initiation, development or acquisition, implementation, operation, support, maintenance, administration, and disposal stages. A promising solution approach to develop airworthiness security guidance based on these constraints is to view physical security threats mostly as an external dependency of airworthiness security, and scope down cyber threats as manageable risks to aircraft safety [9].

Compliance with the necessary certification data and objectives of the guidance for aircraft can be achieved either by a blended process that integrates safety with security, or through a segregated process that defines a separate security process from the safety process. A major future challenge is to sufficiently integrate safety and security processes to maintain overall consistency as well as ensure that outputs of safety process include all hazards identified by security analyses.

VIII. CONCLUDING REMARKS

In the next two decades and beyond, civil aviation will undergo a significant transformation at airports and in airspace to more safely, securely, efficiently accommodate the predicted multifold increase in air traffic, challenging "green" aviation goals, aging traveler population, and globalization trends. This paper envisioned the modernized air transport system to be a net-centric operations environment with commercial aviation, general aviation, unmanned aircraft operators, and other stakeholders on the ground, worldwide. The future "e-enabled aircraft" plays a central role by tightly integrating with ground systems and neighbor airspace users. This enables seamless operations on user-preferred global routes while exceeding future airspace system performance demands and offering a "home/office" experience to travelers. We identified some major cyber–physical security risks with near-term and longer term capabilities for controlling air traffic, i.e., ADS-B, and managing aircraft operations, i.e., digital content refresh and distribution over broadband Internet connectivity. We described the critical role of cyber–physical security in bridging the potentially growing gap between economically driven technology advances and safety driven regulations and standards. Additionally, the paper discussed some major challenges and offered unique perspectives in the future evolution of IP ATN. ■

APPENDIX I

Table 1. Common Abbreviations in this Paper

Abbreviation	Description
3-D / 4-D	3 / 4 dimensional
ADS-B	Automatic Dependent Surveillance Broadcast
ARINC	Aeronautical Radio Incorporated
ATM	Air Traffic Management
ATN	Aeronautical Telecommunication Network
FAA	Federal Aviation Administration
GPS	Global Positioning System
ICAO	International Civil Aviation Organization
IPv4/v6	Internet Protocol version 4/version 6
RFID	Radio Frequency Identification
RTCA DO	Radio Technical Commission for Aeronautics Design Objective
UAT	Universal Access Transceiver
UHF/VHF	Ultra/Very High Frequency
VDL	VHF Data Link

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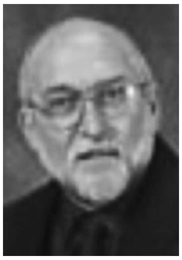

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